each of those characteristics simplifies programming tasks and improves developer productivity. **(8 marks)**

QUESTION TWO

Imagine you're leading a team of programmers who are developing a new web application for an e-commerce platform. The team consists of developers with varying levels of experience and expertise in programming languages like JavaScript, Python, Java, SQL, and Ruby. As the project progresses, you notice that each developer has their own preferred programming approach to solving problems. Some developers advocate for imperative programming approach, others prefer an object-oriented programming (OOP) approach, while others advocate for declarative programming approach. As the lead programmer:

a. Describe to the various teams the characteristic and concepts of each programming approach. **(12 marks)**

b. In your own opinion, which programming approach do you think is most suitable for developing the web application. Give reasons for your answer. **(8 marks)**

QUESTION THREE

You've been tasked with developing a software solution for a small retail business that wants to manage its inventory and sales. The system should track products, handle sales transactions, and generate reports on inventory levels, sales performance, and revenue.

a. Compare and contrast how the structured program design approach and the object-oriented program design approach would address the requirements of the retail inventory and sales management system. Consider factors such as code organization, scalability, maintenance, and ease of implementation. **(12 marks)**

b. Discuss the advantages and disadvantages of each approach in this context and justify your choice of approach for this project. **(8 marks)**

a. Using a diagram, describe the structure of a python program. **(5 marks)**

b. Identifiers are common elements of python programs.

  i. Define an identifier **(1 mark)**

  ii. List down the rules for naming identifiers **(4 marks)**

c. Python programs can be executed using conditional and repetition statements. Using examples or illustrations, describe the conditional and repetition statements used in python programs. **(10 marks)**

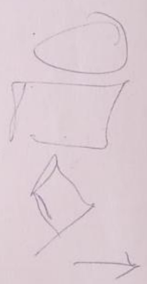
End.

QUESTION FOUR

a. Differentiate between an algorithm and a program flow chart. (**2 marks**)

b. Draw and explain the operation of any three program flow charts symbols. (**9 marks**)

c. Design a program flow chart representation for the logical statements below used by a certain tertiary institution to grade the student's results. (**9marks**).



i. **OPEN RESULT FILE** FOR INPUT OF STUDENTS DETAILS.
ii. **READ** FIRST STUDENT'S result details
iii. **ARE THERE** MORE Results in the **RESULTS FILE**?
iv. **IF NO** MORE RECORDS THEN, **CLOSE RESULT** FILE, and **END**
v. **ELSE**
vi. **IS** Result > 80 THEN, **GRADE = A**
vii. **ELSE**
viii. **IS** Result > 70 THEN, **GRADE = B**
ix. **ELSE**
x. IS Result > 60 THEN, **GRADE = C**
xi. **ELSE**
xii. **IS** Result > 50 THEN, **GRADE = D**
xiii. **ELSE**
xiv. Result = **FAILURE**
xv. **PRINT** STUDENT RESULTS AND READ NEXT Result IN THE **RESULT FILE**

QUESTION FIVE

Python is a general-purpose programming language used in almost every domain raging from web development, game development to artificial intelligence, machine learning, data analytics and natural language processing. Because of this many programmers have adopted a structure of a python program.

4

MAKERERE UNIVERSITY BUSINESS SCHOOL
END OF SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF BUSINESS COMPUTING (BBC)
OF MAKERERE UNIVERSITY, ACADEMIC YEAR 2023/ 2024

COURSE NAME: Programming Theory and Problem Solving
COURSE CODE: BUC1223
YEAR OF STUDY: One **SEMESTER:** Two
DATE: Wednesday 8th May 2024 **TIME:** 9am-12.00Noon

INSTRUCTIONS

i. This examination is a theory examination. The practical part has been done as a semester project.
ii. This examination contributes **60 marks** and the project **40 marks**.
iii. Attempt **ANY THREE** Questions. All questions carry equal marks
iv. The final mark will consist of the project mark and the marks obtained from this examination.
v. **Do NOT** write anything on the question paper

QUESTION ONE

a) The Computer Program Development Lifecycle refers to the process of designing, creating, deploying, and maintaining computer programs. It encompasses a series of stages or phases that guide developers through the entire lifecycle of program development. Describe the stages involved in the computer program development lifecycle by mentioning the objective and activities of each stage. **(12 marks)**

b) Programming in high level programming languages comes with a number of characteristics that make programming even easier for the programmer today. These characteristics contribute to their ease of use and efficiency in program development. In light of the above statement, identify and explain those characteristics highlighting how